

# **To What Extent Can AI Tools Accurately Diagnose Common Diseases in Low-Resource Settings Like Rural India, and What Biological or Technical Factors Limit Their Effectiveness?**

Anaya Shah

## **ABSTRACT**

Artificial intelligence has been revolutionizing the healthcare sector across the globe. Its application in rural areas is often a debated topic. India is a global leader when it comes to pharmaceuticals and development of technologies yet a vast majority of the population that lives in rural areas lack basic health facilities. Artificial intelligence is a revolutionary chance to solve the healthcare inequality in India in rural areas. The given study measures the diagnostic effectiveness of AI when it is used in the resource-limited setting, namely, the biological and technical limitations that undermine its functioning. The paper demonstrates a considerable gap in performance by exploring core technologies (deep learning and NLP) and analyzing evidence in high-burden diseases such as TB, malaria, and oncology. As case studies in organizations such as Qure.ai and Niramai show, although AI is effective in specific visual tasks, there is a lack of reliability due to genetic diversity, co-morbidities, dataset bias, and infrastructure shortage. The researchers conclude that AI can best be used as a clinical support tool. India needs to pursue a combined approach of algorithmic innovation and strong capacity to establish public health infrastructure as well as localized governance of data to attain genuine health equity.

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## 1. ARTIFICIAL INTELLIGENCE IN HEALTHCARE

Artificial Intelligence are computer programs that mimic human intelligence like solving problems, analyzing, learning and reasoning. Artificial Intelligence came to exist around the 1950s .Over the past few years the field of artificial intelligence has experienced massive growth and evolution across all sectors. Healthcare system has also experienced a revolution through AI in the past few years specially because current healthcare system involves large and various types of data information such as molecular and genetics profiles, patient records and real time health monitoring which requires modern and refined AI algorithms (Topol, 2019) .

### *Use of AI healthcare involves-*

**Image analysis:** AI models can evaluate, study, recognize patterns in medical images like X-Rays and MRI scans etc. , and help identify disease and their symptoms (Topol, 2019) .

**Medical guidance:** AI can also assist in analysis of the records of patients and predict effective treatment options (Topol, 2019) .

**Drug Discovery:** AI, especially through its in-silico methods has made the entire pipeline of identifying potential drug candidates amongst several molecules, efficient and faster (Mak et al., 2024).

**Diagnostic forecasting:** By taking a look at population molecular and genetic information, recognizing patterns and analyzing patient records, AI can also predict if a patient or the population possess a health threat (Khalifa & Albadawy, 2024).

**Automating administrative work:** AI through it subsets like Natural Language Processing and Machine learning, makes it easier to streamline data related administrative work from hand written data to digitally stored data (Topol, 2019).

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## 2. INADEQUATE HEALTHCARE SYSTEM IN RURAL AREAS

A paradox exists in the Indian Healthcare system. On one end India is a global leader when it comes to manufacturing pharmaceuticals and developing technologies but on the other end it fails to meet the demand of the vast population living in rural areas (Nair et al., 2022). India has 20.6 healthcare worker per 10000 people which is far below the prescribed standard of 44.5 healthcare workers per 10000 by WHO (Mehta et al., 2024). India faces a shortage of healthcare workers and its rural areas remain highly underserved. 65% of Indian population lives in rural areas (Arora, 2024). India has 15.9 beds per 10000 which is again far less than the prescribe standard by WHO. There is a major shortage of medical professional in rural areas. The Rural Health Statistics of India (2021) reveals that there is a shortage of 79.5% shortfall of specialists at the Community Health Centres (CHCs) (A. Joshi et al., 2024).

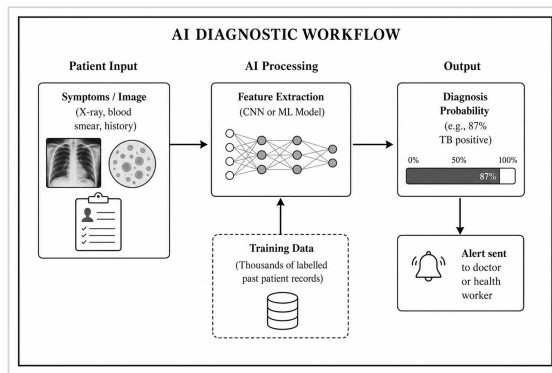
This results in an alarming situation in rural areas. Diseases which can usually be prevented from happening like tuberculosis, malaria, dengue, and diarrhoea are a common occurrence in rural areas because of poor sanitation, poor living conditions and lack of adequate healthcare (Narain et al., 2023). Due to changes in the lifestyle and food nowadays diabetes and heart related issues have been on a rise as well (Swaminathan et al., 2017). Cancers like cervical cancer, breast cancer, and oral cancer often go unnoticed until the later stages because there is a lack of preventative care (Tripathi et al., 2014). Many deaths can be prevented if diagnosed on time but rural areas suffer from diagnosis delays because of their insufficient healthcare facilities.

It because of this fact that even under such heavy health burden in rural areas which still remain understaffed when it comes to their healthcare system that Artificial Intelligence has become a subject of high hopes and interest in order to revolutionize

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the Healthcare system in the rural areas. India in the past few years has faced a massive digital revolution (Rana et al., 2024). One can find smartphones almost everywhere nowadays but digital illiteracy and lack of proper electric supply in rural areas raises a question of till what extent can AI be of help in rural setting.

### 3. AI MECHANISMS FOR DIAGNOSIS OF DISEASES

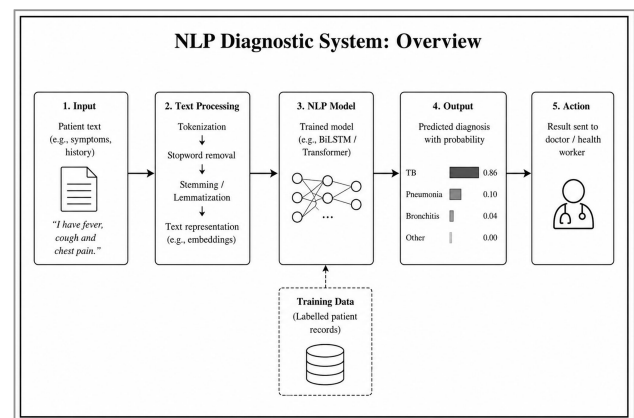


*Figure 1 AI in Diagnosis Workflow*

#### Symptom checkers

Platforms like Ada Health, Babylon Health and India-specific platforms like mFine and Practo are AI based and are very useful when it comes to disease diagnosis

(Gilbert et al., 2020). These are usually in app format or web based chatbots that to which users can describe their symptoms to. Through Natural Language Processing (NLP), these platforms can understand human speech. These platforms then identify patterns and recognizes important symptoms by comparing the user data to other medical data in their medical library (Nadkarni et al., 2011). The platform then diagnoses the disease or potential disease. As effective as they may be for the diagnosis of several common diseases like respiratory diseases, etc., they are not very accurate when it comes to complex or overlapping symptoms across multiple diseases (Leaman et al., 2015).



*Figure 2 NLP Mechanism for diagnosis*

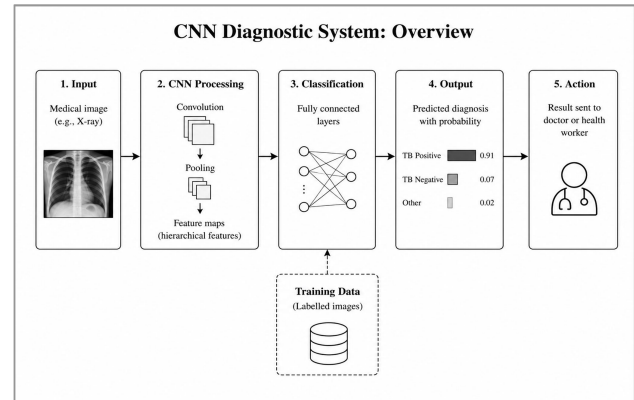
## Image Diagnosis

Convolutional Neural Networks (CNNs) is a subset of AI that is capable of studying, finding patterns and learning from medical images like:

- X-rays
- Microscopy images
- Retinal scans
- Skin photographs

CNN mimics the way our brain understands visual information. Image based diagnosis is as of current the most accurate as well as well validated in terms of AI and its application in healthcare (Sarvamangala & Kulkarni, 2022). It works by not simply looking at the image as a whole but pixel by pixel instead. It converts each pixel into numeric values and forms a numerical grid. The AI model then starts its training cycle by recognizing visual markers by analysing brightness, shadows, lines and contrasting regions of the images. With every training cycle the AI model attempts to improve its accuracy.

The final model when fed with an unfamiliar image, based on its training gives out a probability score for the disease. CNNs are used to detect incidence of TB in X-rays (Showkatian et al., 2022).



*Figure 3 CNN Mechanism for diagnosis*

## AI wearable sensor devices

In several rural places, hospitals are often located at a far distance from the homes of the rural people, making travel a hassle. AI wearable sensor devices are a game changer in cases like these as not only are they easy to wear, they help monitor a person's health 24/7, help diagnose disease in early stages and within their homes

itself without having to far located hospitals (Shajari et al., 2023).

*Some of these devices are:*

- **Pulse oximeters:** These devices constantly monitor blood oxygen levels of the patient (Cabanias et al., 2024). It detects if there is a sudden drop in the blood oxygen level and can alert for potential heart diseases or pneumonia.
- **Portable ECG patches:** These are wearable patches that record the activity of heart for days. The AI algorithm detects if any abnormal activity occurs like irregular heartbeats which could lead to a stroke (Wolf et al., 1991).
- **Glucometers with AI integration:** These devices monitor blood glucose levels and use AI's predictive algorithms to warn the patients early of chances of their sugar level dropping low, which helps patient manage diabetes better.

- **Acoustic sensors (digital stethoscopes):** These are like digital stethoscopes that can record lung sounds and then using AI algorithms, analyse them to diagnose chances of respiratory diseases like tuberculosis or pneumonia.

Typically, people would travel to hospitals or testing labs for getting medically tested. Point of Care Testing with AI algorithms integrated makes medical testing and diagnosing possible within the patient's home itself or at a local nearby pharmacy store (Rajput et al., 2024). For example, a healthcare worker can simply place a pulse oximeter on a patient's finger and AI algorithms can check if something is wrong.

### **Federated Learning**

AI models need data on which they can be trained in order for AI algorithms to be able to study, analyse and predict something. Federated learning makes it possible for data to be shared across

several healthcare institutes while ensure privacy of a patient's medical records is preserved. Federated learning allows AI models to train locally and only share the data converted into mathematical values instead of raw data of the patient's medical records to different locations (Teo et al., 2024).

#### 4. THE ACCURACY OF AI TOOLS

Accuracy of AI tools is judged by looking at parameters (Rainio et al., 2024) like:

*Table 1 Parameters for AI model evaluation*

| Measure            | Definition                                                | Importance                                                                        |
|--------------------|-----------------------------------------------------------|-----------------------------------------------------------------------------------|
| <b>Sensitivity</b> | The number of patients that AI recognises to be diseased. | High sensitivity is important as it ensures no patient's disease goes undetected. |

|                                        |                                                                                 |                                                                                       |
|----------------------------------------|---------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| <b>Specificity</b>                     | The number of patients that AI recognises to be healthy.                        | High specificity because of false positives in diagnosis causes anxiety and distress. |
| <b>AUC (Area Under the Curve)</b>      | A score ranging from 0 to 1.0. Scores above 0.90 are considered as good scores. | An overall score that summarises everything.                                          |
| <b>Positive Predictive Value (PPV)</b> | A value that indicates how truly positive is an AI positive result              | Important for rare diseases where false positives are often seen                      |

|                                        |                                                                    |                                        |
|----------------------------------------|--------------------------------------------------------------------|----------------------------------------|
| <b>Negative Predictive Value (NPV)</b> | A value that indicates how truly negative is an AI negative result | Important to validate negative results |
|----------------------------------------|--------------------------------------------------------------------|----------------------------------------|

### Tuberculosis (TB)

In a person with Tuberculosis the tissues of the lungs suffer damage. This damage can be viewed using medical imaging like X-Ray. CNNs are excellent AI models when it comes to learning patterns and identifying visual markers in medical images .

*AI has proven to effective in TB detection:*

- CAD4TB is a software developed by Delft Imaging which is highly effective with a sensitivity of 87% to 91% (Qin et al., 2019).
- qXR is an AI tool developed in India by Qure.ai. According to a study published in *Scientific Reports*, its

AUC was approximately 0.94 (Maka Shrestha et al., 2024).

- In a study published in the *The Lancet Digital Health* in the year 2020, it was found that AI models are better when compared to non-specialist doctors when it comes to reading chest X-rays in three countries, including India (Qin et al., 2021).

These mentioned studies were performed in urban hospitals with high-quality technology for medical imaging. In rural setup where the equipment quality for medical imaging in hospitals is comparatively lower than the urban areas and a shortage of skilled technicians, the accuracy scores naturally drop by 10-20% (Melendez et al., 2016).

### Malaria:

Every year across the globe around 600,000 lives are claimed due to malaria . India contributes to this statistic significantly as well. In order to combat this, it is important to identify the parasite



causing microorganisms like *Plasmodium falciparum* and *Plasmodium vivax*. This in turn helps in identifying what kind of Malaria it is i.e whether it is the deadly kind or the one that relapses. For this identification blood smears need to be studied under microscope by trained specialists.

*AI offers solutions to this such as:*

- An NIH-funded AI model showed a sensitivity of approximately **98.1%** and specificity of **99.2%** in identifying *Plasmodium falciparum* in **thick blood smear images**. This is comparable to the accuracy of specialists in recognizing the same (Liu et al., 2023).
- UC Berkeley developed a device called CellScope which can attach itself to the back of the smartphone and act as a microscope (Breslauer et al., 2009). This allows health workers to visualize and identify malaria parasites with assistance of AI without having to

depend on a specialist or labs (Maturana et al., 2023).

The limitation of AI in Malaria detection is the it is heavily dependent on the input given to AI which are blood smears . The accuracy of result depends upon how well the blood smears are prepared which ultimately depend upon the skills of the person responsible for making it. Therefore Rapid Diagnostic Tests are still considered more reliable than the AI based tools (Maturana et al., 2023).

### **Diabetic Retinopathy**

Over 77 million people in India are known to have Type 2 diabetes (Pradeepa & Mohan, 2021). Among these, 17% have diabetic retinopathy. It is disease that causes blindness which is irreversible in diabetic patients (Rema et al., 2005). Happens due to damage in the blood vessels retina. It can be prevented, provided that it gets diagnosed early. In order for that to happen, regular check-ups

by trained ophthalmologists is required. They are rare in Rural India .

*AI's role in Diabetic Retinopathy diagnosis is as follows:*

- **Google Research developed an AI model that is able to diagnose** diabetic retinopathy by analysing images of retinal fundus (Bahl & Rao, 2022). It has a sensitivity of 97.5% and specificity of 93.4% (Abràmoff et al., 2018). This is comparable to the ophthalmologist level of accuracy.
- Google Research also developed another AI model in collaboration with the **Aravind Eye Hospital** in Tamil Nadu. It analyses images and detect diabetic retinopathy with high accuracy. This showcases the potential of accurate screening of the disease even in settings lacking proper resources .
- In a study which was conducted in Southern India, over 3000 diabetic patients were screened using an AI

model for image analysis. Automated AI models like these not only help reduce time in screening and detection but also reduce dependence on skilled ophthalmologists that are rare in rural areas. Faster screening allows faster treatment and damage prevention (Abràmoff et al., 2018).

The application of these AI models is dependent on digital fundus camera which can cost up to 2-3 lakh rupees approximately. These cameras need to be properly maintained and timely calibrated. This become a limitation for the application of these AI models in a rural setting.

## **Skin Diseases**

Skin diseases are highly common in rural India (Hay et al., 2014). These include conditions such as:

- **Leprosy** (India ranks 2<sup>nd</sup> highest in occurrence of Leprosy globally)
- **Fungal skin infections**

- **Scabies**
- Parasitic infections
- **Skin cancer** (It is preventable is diagnosed on time)

*AI tools for its detection:*

- In a study published in *Nature* it was seen that a CNN model given 130,000 clinical images as training data was able to diagnose and classify skin cancer with high sensitivity and specificity, comparable to skilled dermatologist-level accuracy (Esteva et al., 2017).
- A study showed that AI models for image analysis can diagnose and classify leprosy by analysing images of skin lesions (Fernandes et al., 2023).

These AI models are mostly tested on people with a lighter skin tone. The AI models struggle in diagnosing with high accuracy in dark skinned people patients which is a common characteristic in Indian population. It also struggles with early

stage detection and cases that are borderline. It is due to these reasons that the application of AI tools is heavily limited in India (Adamson & Smith, 2018).

### **Cardiovascular Disease and ECG Analysis**

28% of all deaths that happen in India are due to cardiovascular diseases (Prabhakaran et al., 2018). These deaths can be prevented if cardiovascular diseases are diagnosed early.

*AI tools for diagnosis of Cardiovascular diseases:*

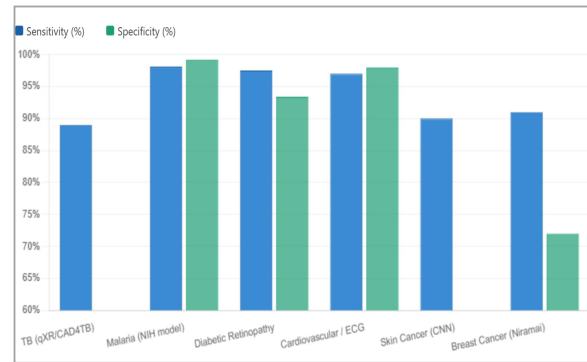
- KardiaMobile is a an ECG device that is portable and can be connected to Smartphones. It is developed by AliveCor. Its Ai algorithm can detect atrial fibrillation with sensitivity of 97% and specificity of 98% (Bumgarner et al., 2018).
- A study published in *Nature Medicine* showed an AI model

capable of detecting and classifying cardiac arrhythmias by analysing ECG data (Khan et al., 2026).

The use of these portable ECG devices is being promoted by several Indian hospitals. This initiative is also supported by ICMR. These AI models are validated in clinical setting. In a real world setting the ECG recordings may get affected due to interference from nearby electric sources, movement of the patient and poor placement of the electrode.

### **Pneumonia**

India also witnesses major incidence of pneumonia, especially in the rural areas (Pandey & Galvani, 2020). CheXNet is an AI model that uses CNN to detect pneumonia . This model struggles with detection in children less than two years old or toddlers (Godbole et al., 2024).



*Figure 4. Sensitivity and Specificity scores of AI models across common diseases*

## **5. ADVANTAGES OF AI DIAGNOSTIC TOOLS IN RURAL SETTINGS**

The benefits of AI and its applications are not limited to diagnostic applications only. AI offers some really compelling advantages like:

### **AI tools are cost effective**

Expanding medical services and hiring several different skilled healthcare workers for varied different tasks is a costly affair. AI on the other hand, once tested and validated can be shared virtually and

easily. Once the AI app has been created, the cost of sending it to just one more person is zero and no extra effort is required (Ugwu et al., 2026a).

Application like Qure.ai would cost around 50-100 rupees per X-ray analysis whereas an expert radiologist would charge 500-2000 rupees for the same because they also have to factor in salary, infrastructure cost, electricity cost as well as cost of equipments (Powles & Hodson, 2017). AI platforms like Niramai's AI 10 times cheaper than the price of a mammogram. Platforms like iKure that have AI diagnosis integrated into them charge around 100 rupees for a consultation which is far less than what a specialist in the urban areas would charge (Xie et al., 2020).

### **Accessibility**

Even in places with poor diagnostic infrastructure, diagnosis can be carried out in homes or local pharmacy stores in rural areas as long as even one health worker

has a portable camera, smartphone and an AI application trained in diagnosis of several diseases specially the one commonly seen in rural areas like TB, Malaria, diabetic retinopathy, pneumonia and skin diseases (Ugwu et al., 2026b).

India has more than 1 million ASHA (Accredited Social Health Activist) health workers. These are community health workers who are mostly women. They may not possess clinical expertise but they are trusted by their community and are able to be physically present in remote areas as well. AI devices that are simple in design, easy to navigate and available in several local languages would help these ASHA network workers to provide expert diagnosis despite their limited clinical expertise (Mohsin et al., 2023).

### **Speed**

Several diseases can be prevented if diagnosed early. AI tools like Qure.ai's Qxr, Google's AI for diabetic Retinopathy and Malaria detection AI can easily

diagnose the disease from looking at the images in under 60 seconds to 5 mins (Wongsrichanalai et al., 2007). This is way faster when compared to the time taken by busy laboratories and skilled technicians. They can take up days or at times months to report results (Negi & Kaushik, 2025).

### **Standardisation**

All healthcare workers are humans at the end of the day and like every other human they get exhausted as well. Health technicians like Radiologists end up disagreeing on their diagnosis about 20% of the times. Unlike humans AI tools don't get exhausted and no matter how many times they run diagnostic analysis the results are always consistent and unbiased . It is due to this that AI helps provide standardised medical care across several districts (Negi & Kaushik, 2025).

### **Scalability**

Attempts are being made to integrating AI diagnostic tools into National Health

programmes. The National TB Elimination programme has already integrated AI into its operations and is using it to scan X-rays across several districts (Ish et al., 2026). Through the Ayushman Bharat Digital Mission (ABDM), the government of India is linking patient's data and AI in a digital information storage system which attempt to connect patients and their medical data to healthcare workers across the country.

## **6. LIMITATIONS OF AI IN HEALTHCARE**

The effective use of AI in Rural areas in being limited by some biological and some technical factors.

### **Biological limitations**

The biology of human body is such that it is unpredictable in several aspects. AI models train on what is considered a standard design for several diseases. But diseases come in varied form and not in a standardized manner. For example

Extrapulmonary TB isn't visible in X-rays. Miliary TB appears like seeds in medical imaging and acts can confuse the AI models at times because they resemble other conditions. Reactivated TB produce varied X-rays. This is called as generalisation failure. In HIV patients that also have TB the X-ray Patterns for Tb can be unusual because of altered immune response (caused due to HIV). In such cases sometimes abnormalities go undetected despite the disease being active (Scott et al., 2025). It is a concern because in India around 100,000 people have both HIV and TB .

The world is full of genetic diversity. For example, Indians can get diabetes at weights lower than Europeans. Therefore, AI trained in the West may not always be accurate for Indian patients, as it may classify an Indian patient as safe even when they might be at risk . Even in India an AI model trained in Delhi may not necessarily be equally accurate for a person in Odisha where genetic profiles

and the diet patterns differ (“Appropriate Body-Mass Index for Asian Populations and Its Implications for Policy and Intervention Strategies,” 2004).

In rural areas patients are rarely affected with a single isolated disease. People in rural areas often have multiple infections because of several factors such as malnutrition, poor sanitation and poverty. TB and Malaria show fever, weight loss and fatigue as common symptoms. These overlapping symptoms serve as noise for AI models and this can result in false positives (Brown & Davis, 2025).

Sometimes the medical images are of poor qualities due to lacking infrastructure and low quality tools. Blood smears may not be properly prepared due to lack of skills. In rural areas a lot of social stigma exists which makes a person not reveal all their symptoms freely to the AI diagnostic apps. Due to poor literacy some rural people may not understand several medical terms.

When inputs are poor then the outputs to become poor naturally.

### **Technical limitations**

#### ***Lack of Diverse Training Data***

AI models tend to have a bias from the data they get trained upon. An AI trained in wealthy Western hospitals or urban cities like Delhi or Mumbai would not be equally accurate for a malnourished person from rural area. AI models trained on lighter skin tones would not be equally effective for darker skin tones even though they give excellent results in the place where they were tested and validated . A vast majority of Indian have medium to dark skinned tone so this created a major concern about the accuracy of AI models developed and trained on Western land and people (Dowie, 2025).

#### ***Lack of Reasoning***

An AI model may give an output saying that there is 87% probability of TB but will be unable to explain the reasons

behind the diagnosis. Health workers who do not understand the technicalities of AI diagnosis will be unable to evaluate AI outputs or even detect if AI makes an error on rare occasions. Doctors as well as patients fail to trust things that they do not understand the complete reasoning for (Kazemzadeh et al., 2024).

#### ***Poor Infrastructure***

Some rural areas have unreliable electricity and electric surges can sometimes destroy expensive electric equipment. Remote areas often have patchy 4G network and some have 2G signals which is not sufficient to upload large sized medical images. AI models also depend on Cloud based processing.

Since AI models specially which are CNN based analyse images to diagnose, they require high quality medical images for which high quality equipment are required (Clement David-Olawade et al., 2025). Most equipment in rural areas are of poor quality and outdated. Many Primary



Healthcare centres in rural areas don't even have basic X-Ray machines. There is also of shortage of trained healthcare staff who would not how to handle this equipment efficiently. There is also a lack of maintenance which often leads to equipment's quality downgrading over time.

There is a concept called Model Drift which says that AI accuracy drops over time. For example, an AI model trained before 2020 wouldn't be able to distinguish the lung damage caused by TB from a COVID-19 caused lung damage. Therefore, these models need to be updated regularly, which is difficult in rural areas (Lacson et al., 2022).

Indian laws are still behind on catching up to AI. Until laws have been defined which define the standards of what AI can be used and not and unless there are laws laid out that protect patient's sensitive data from being leaked, AI use will not be fully accepted. The Central Drugs Standard

Control Organisation (CDSCO) is working on creating guidelines for AI and ML based healthcare tools under the Medical Devices Rules, 2017, but it is still a work in progress (U. Joshi et al., 2026).

## 7. THE QURE.AI CASE

Qure.ai is an AI healthcare firm which is based in Mumbai. It has created an AI system called qXR. It can identify abnormalities in chest X-rays and diagnose TB and other lung related diseases. It was deployed in government hospitals and primary health centres in India, Nigeria and some other countries.

qXR was used to screen thousands of patients in under-served districts of Maharashtra and Uttar Pradesh in a pilot project in collaboration with USAID and the Indian government. It was found that that the AI was as accurate as senior radiologists and it also shortened the time taken to diagnose TB by several days

down to less than a minute. But, the tool relied on X-ray machines which were not available in the most remote locations. Therefore, it could only reach semi-urban primary health centres and was not of help to the under-served villages (Vijayan et al., 2023).

## **8. NIRAMAI CASE**

Niramai (Non-invasive Risk Assessment with Machine Intelligence) is a start-up that is based in Bangalore. It involves the use of a thermal scan integrated with AI in order to diagnose breast cancer without the use of costly mammography devices or specialized radiologists. It requires a cheap thermal camera which is affordable even in the rural areas. The camera is required to identify the presence of heat due to the growth of tumours. The thermal images are analysed by AI and if suspicious patterns exist then they are detected and identified. One huge benefit is that it is non-invasive and does not need radiation.

Its use has been implemented in rural health camps in Karnataka and Tamil Nadu. The AI was able to achieve a sensitivity of 91% and specificity of 72% in initial studies. The specificity still needs to be improved to avoid unwarranted anxiety of patients and subsequent costs (Adapa et al., 2025).

## **9. MICROSOFT AI FOR HEALTH CASE**

Microsoft in its initiative of AI for health along with Indian Research institutes funds a project that is focused on automating the detection of malaria parasites in blood smears with microscopes that can be mounted on smartphones (Nema et al., 2022). Regions like Odisha and Jharkhand which have the frequent occurrence of malaria but the lowest number of microscopists are being focused upon. The model was trained on several blood smear images of Indian patients. The accuracy turned out to be comparable to expert

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microscopists. But the results are still limited to the quality of blood smear preparation and image quality.

## 10. ETHICAL CONCERNS

Widespread use of AI raises several concerns such as:

### **Data Privacy**

AI models need lots of sensitive data of patients to be trained. In rural areas where patients do not understand the risks associated with data leakage, data privacy and informed consent may get exploited (Oduro et al., 2008).

### **Job replacement**

AI taking over human jobs has become a common fear of the era. Many fear that AI may replace doctors but experts believe that AI more like a super assistant instead of a replacement (Rony et al., 2024). It assists experts and making better decisions. AI still cannot do what a human

can and also cannot completely understand human emotions and the struggle of patients and their families. If AI makes a mistakes then whenever it comes to health, the stakes are always too high to accommodate risks.

### **Lack of trust**

People in rural areas often do not trust AI and healthcare system. They rely a lot on traditional practices and household and neighbourhood practices existing in their communities for years. Therefore, as important it is to build new technologies it is also important to ensure that people are well informed and educated about it. AI tools are also not available all local languages which again causes lack of trust (Rony et al., 2024).

## 11. CONCLUSION

AI tools are capable of diagnosing common diseases in rural areas in India. The effectiveness of these AI tools is

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limited by factors such as biological complexity, genetic diversity, data biases, lack of quality equipment and technology, lacking infrastructure, language barriers, socio-economic differences, social stigmas and lack of trust.

Therefore, even if AI is comparable to health specialists when it comes to diagnosing common diseases, it is still not a complete diagnostic solution that is capable of replacing healthcare workers.

AI can be used as a means to strengthen the healthcare system in rural areas while also working on improving infrastructure facilitates, improving healthcare workers' shortage, better outreach programmes, stronger government policies and laws governing AI in order to save lives in underserved rural areas.

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